Online Resource 2

Baseline and re-test details

Supplementary material to: *How the Classical Baseline Shapes Machine-Learning Speed Claims in Topology Optimization.* Shaoliang Yang and Jun Wang, Santa Clara University. Corresponding author: Jun Wang, jwang22@scu.edu. Results in Engineering.

Reading guide. OAT is Optimize Any Topology, a diffusion model; HGTO is the graph-based design method. CPU/GPU, central/graphics processing unit; RAM, random-access memory; GiB/MiB, gibibytes/mebibytes; CUDA, NVIDIA's GPU computing platform; SIMP, solid isotropic material with penalization; OC, optimality criteria; PCG/CG, preconditioned/conjugate gradient; MBB, Messerschmitt–Bölkow–Blohm beam; DOF, degree of freedom; SPD, symmetric positive definite. P1 is the cantilever and P2 the MBB half-beam. Solver checks compare the element stiffness matrix, displacement and physical density; diagnostic tests are displayed separately from the tests required before timing. Software and file names identify the implementations and records needed to reproduce the checks. PARDISO and SuperLU are sparse direct equation solvers (the CPU alternative and the reference); top88 is the 88-line classical topology-optimization code; KE is the element stiffness matrix; "Ref" is OAT's refinement step, which our harness does not run; "its" abbreviates iterations; P-cores are the CPU's performance cores; H100 is the NVIDIA GPU on which OAT's offline cost was reported.

1. Environment

Captured 2026-09-24T14:15:33+00:00 (`2026-09-24_env.json`). Operating system Windows-11-10.0.26200-SP0; CPU Intel(R) Core(TM) i5-14600KF (14 cores, 20 threads); 95.8 GiB RAM; GPU NVIDIA GeForce RTX 5090 (32607 MiB, driver 616.56). Two Python environments were used: environment A for the GPU-solver tests and environment B for the PARDISO tests (the interpreters recorded in the two test files).

Environment	Python	NumPy	SciPy	PyTorch	CUDA	pypardiso
A	3.12.13	2.4.3	1.17.1	2.11.0+cu128	12.8	–
B	3.10.18	2.2.6	1.15.3	2.5.1	12.1	0.4.7

2. Solver correctness

All 44 tests of `baseline_tests_gpu.json` and `baseline_tests_pardiso.json` are listed. 40 of 40 gating tests passed; gating tests had to pass before any timing. Four diagnostic tests were not required to pass; three recorded values outside their diagnostic limits and are shown as measured.

Solver	Test	Value	Limit	Pass	Gating
GPU	KE symmetry max|KE-KE^T|	2.78×10^{-17}	<1e-14	yes	yes
GPU	KE: 3 zero eigenvalues (max |lambda_1..3|)	1.71×10^{-16}	<1e-12	yes	yes
GPU	KE: 5 positive eigenvalues (min lambda_4..8)	0.495	>1e-3	yes	yes
GPU	KE: rigid rotation in nullspace |KE r|	2.78×10^{-17}	<1e-13	yes	yes
GPU	KE equals top88 closed form	1.67×10^{-16}	<1e-14	yes	yes
GPU	patch test: interior forces under linear field	5.90×10^{-17}	<1e-14	yes	yes
GPU	P1 cantilever n=32: GPU eager vs eager (nondeterminism floor), max |dxPhys|	5.77×10^{-8}	recorded; must be <1e-6	yes	yes
GPU	P1 cantilever n=32: GPU loop graph vs eager, max rel. compliance diff (40 its)	8.14×10^{-13}	<1e-10	yes	yes
GPU	P1 cantilever n=32: GPU loop graph vs eager, max |dxPhys|	1.07×10^{-7}	<1e-6 (floor-calibrated)	yes	yes
GPU	P1 cantilever n=32: GPU vs CPU top88-faithful port, rel. compliance diff at it 40	2.50×10^{-10}	<1e-5	yes	yes
GPU	P1 cantilever n=32: GPU vs CPU top88-faithful port, max |dxPhys| at it 40	4.02×10^{-7}	<1e-3	yes	yes
GPU	P2 MBB half-beam n=48: GPU eager vs eager (nondeterminism floor), max |dxPhys|	4.99×10^{-10}	recorded; must be <1e-6	yes	yes
GPU	P2 MBB half-beam n=48: GPU loop graph vs eager, max rel. compliance diff (40 its)	1.35×10^{-12}	<1e-10	yes	yes
GPU	P2 MBB half-beam n=48: GPU loop graph vs eager, max |dxPhys|	1.04×10^{-9}	<1e-6 (floor-calibrated)	yes	yes

Solver	Test	Value	Limit	Pass	Gating
GPU	P2 MBB half-beam n=48: GPU vs CPU top88-faithful port, rel. compliance diff at it 40	2.26×10^{-11}	<1e-5	yes	yes
GPU	P2 MBB half-beam n=48: GPU vs CPU top88-faithful port, max |dxPhys| at it 40	1.10×10^{-7}	<1e-3	yes	yes
GPU	P1 cantilever n=32: restriction = adjoint of prolongation	3.55×10^{-16}	<1e-12	yes	yes
GPU	P1 cantilever n=32: element-local Galerkin = global P^T K P	2.40×10^{-16}	<1e-12	yes	yes
GPU	P1 cantilever n=32 [uniform]: PCG vs sparse direct, rel. error in u	1.00×10^{-10}	<1e-6	yes	yes
GPU	P1 cantilever n=32 [uniform]: PCG true residual without fallback	2.75×10^{-9}	<=1e-7	yes	yes
GPU	P1 cantilever n=32 [random, without voids]: PCG vs sparse direct, rel. error in u	2.52×10^{-10}	<1e-6	yes	yes
GPU	P1 cantilever n=32 [random, without voids]: PCG true residual without fallback	7.93×10^{-9}	<=1e-7	yes	yes
GPU	P1 cantilever n=32 [SIMP design at iteration 60]: PCG vs sparse direct, rel. error in u	6.87×10^{-8}	<1e-6	yes	yes
GPU	P1 cantilever n=32 [SIMP design at iteration 60]: PCG true residual without fallback	8.52×10^{-9}	<=1e-7	yes	yes
GPU	P1 cantilever n=32 [SIMP design at iteration 60]: CUDA-graph path equals eager path	3.12×10^{-15}	<1e-12	yes	yes
GPU	P1 cantilever n=32 [stress test: 30 % isolated voids]: guarded solve true residual	9.88×10^{-9}	<=1e-7	yes	no
GPU	P2 MBB half-beam n=64: restriction = adjoint of prolongation	5.57×10^{-16}	<1e-12	yes	yes
GPU	P2 MBB half-beam n=64: element-local Galerkin = global P^T K P	2.44×10^{-16}	<1e-12	yes	yes
GPU	P2 MBB half-beam n=64 [uniform]: PCG vs sparse direct, rel. error in u	3.38×10^{-11}	<1e-6	yes	yes
GPU	P2 MBB half-beam n=64 [uniform]: PCG true residual without fallback	3.25×10^{-9}	<=1e-7	yes	yes
GPU	P2 MBB half-beam n=64 [random, without voids]: PCG vs sparse direct, rel. error in u	9.60×10^{-11}	<1e-6	yes	yes
GPU	P2 MBB half-beam n=64 [random, without voids]: PCG true residual without fallback	7.14×10^{-9}	<=1e-7	yes	yes
GPU	P2 MBB half-beam n=64 [SIMP design at iteration 60]: PCG vs sparse direct, rel. error in u	2.39×10^{-7}	<1e-6	yes	yes
GPU	P2 MBB half-beam n=64 [SIMP design at iteration 60]: PCG true residual without fallback	4.09×10^{-9}	<=1e-7	yes	yes
GPU	P2 MBB half-beam n=64 [SIMP design at iteration 60]: CUDA-graph path equals eager path	4.22×10^{-14}	<1e-12	yes	yes
GPU	P2 MBB half-beam n=64 [stress test: 30 % isolated voids]: guarded solve true residual	3.23×10^{-5}	<=1e-7	no	no
PARDISO	P1 cantilever n=64 [uniform]: PARDISO (SPD, phase 23) vs SuperLU, rel. difference	1.63×10^{-13}	<1e-8	yes	yes
PARDISO	P1 cantilever n=64 [random, without voids]: PARDISO (SPD, phase 23) vs SuperLU, rel. difference	2.29×10^{-13}	<1e-8	yes	yes
PARDISO	P1 cantilever n=64 [SIMP design at iteration 60]: PARDISO (SPD, phase 23) vs SuperLU, rel. difference	4.03×10^{-13}	<1e-8	yes	yes
PARDISO	P1 cantilever n=64 [stress test: 30 % isolated voids]: PARDISO (SPD, phase 23) vs SuperLU, rel. difference	1.66×10^{-8}	<1e-8	no	no
PARDISO	P2 MBB half-beam n=64 [uniform]: PARDISO (SPD, phase 23) vs SuperLU, rel. difference	5.72×10^{-13}	<1e-8	yes	yes
PARDISO	P2 MBB half-beam n=64 [random, without voids]: PARDISO (SPD, phase 23) vs SuperLU, rel. difference	7.98×10^{-13}	<1e-8	yes	yes
PARDISO	P2 MBB half-beam n=64 [SIMP design at iteration 60]: PARDISO (SPD, phase 23) vs SuperLU, rel. difference	3.36×10^{-13}	<1e-8	yes	yes
PARDISO	P2 MBB half-beam n=64 [stress test: 30 % isolated voids]: PARDISO (SPD, phase 23) vs SuperLU, rel. difference	1.06×10^{-4}	<1e-8	no	no

Agreement with SuperLU after 150 iterations (`baseline_equivalence.json`): relative difference in final compliance and largest absolute difference in physical density.

Problem	Mesh	Solver	Rel. compliance difference	Max. density difference	Pass
P1 cantilever	64^2	gpu	3.46×10^{-9}	3.04×10^{-7}	yes
P1 cantilever	64^2	pardiso	2.37×10^{-13}	4.39×10^{-11}	yes
P1 cantilever	256^2	gpu	4.45×10^{-10}	5.03×10^{-8}	yes
P1 cantilever	256^2	pardiso	3.24×10^{-12}	6.54×10^{-11}	yes

Problem	Mesh	Solver	Rel. compliance difference	Max. density difference	Pass
P2 MBB half-beam	64^2	gpu	8.70×10^{-10}	3.68×10^{-7}	yes
P2 MBB half-beam	64^2	pardiso	9.29×10^{-12}	5.49×10^{-9}	yes
P2 MBB half-beam	256^2	gpu	2.08×10^{-11}	1.30×10^{-7}	yes
P2 MBB half-beam	256^2	pardiso	1.03×10^{-11}	2.57×10^{-9}	yes

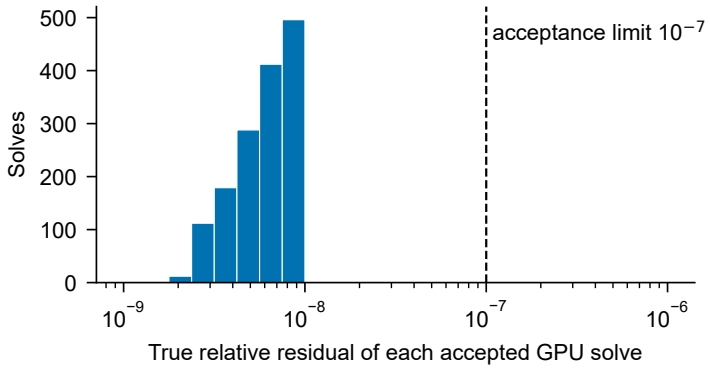


Fig. OR2.1 True relative residual $\|f - Ku\|/\|f\|$ of every accepted GPU solve in the ten pinned 150-iteration benchmark runs (1500 solves; cantilever and MBB beam, 64^2 to 1024^2). 0 solves fell back to the direct solver. The dashed line is the acceptance limit. GPU, graphics processing unit; MBB, Messerschmitt–Bölkow–Blohm beam; **f**, load; **K**, stiffness matrix; **u**, displacement.

3. Scaling of the baseline with mesh size

Problem	Mesh	Solver	Unknowns	ms per SIMP iteration	150 iterations (s)	CG iterations per solve
P1 cantilever	64^2	gpu	8450	13.5	2.01	17.0
P1 cantilever	64^2	pardiso	8450	3.8	0.58	–
P1 cantilever	128^2	gpu	33282	20.0	2.97	17.0
P1 cantilever	256^2	gpu	132098	29.4	4.34	17.0
P1 cantilever	256^2	pardiso	132098	89.7	13.60	–
P1 cantilever	512^2	gpu	526338	146.9	22.40	16.5
P1 cantilever	1024^2	gpu	2101250	286.9	43.33	16.0
P2 MBB half-beam	64^2	gpu	–	13.4	1.98	17.0
P2 MBB half-beam	64^2	pardiso	–	3.9	0.59	–
P2 MBB half-beam	128^2	gpu	–	19.6	2.94	17.0
P2 MBB half-beam	256^2	gpu	–	30.5	4.51	18.0
P2 MBB half-beam	256^2	pardiso	–	93.6	14.11	–
P2 MBB half-beam	512^2	gpu	–	168.7	25.23	17.0
P2 MBB half-beam	1024^2	gpu	–	287.7	44.29	16.0

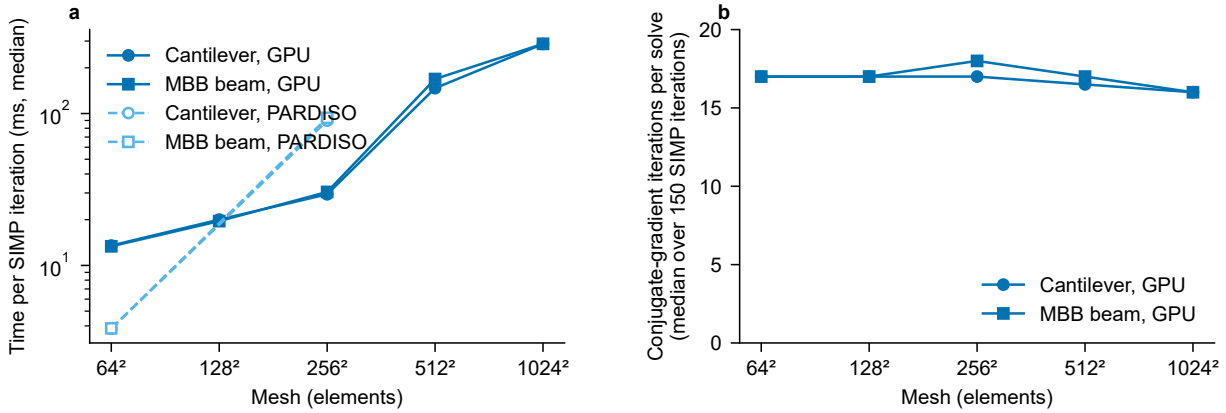


Fig. OR2.2 Scaling of the strong baseline (pinned runs, medians). **a** Time per SIMP iteration for the GPU multigrid solver and PARDISO. **b** Conjugate-gradient iterations per solve of the GPU solver. SIMP, solid isotropic material with penalization; GPU, graphics processing unit. Source: `figures/data/ed3_scaling.csv`.

4. CPU placement

Every timed stage was run twice: under default Windows scheduling and pinned to the six performance cores at high priority. The paper reports only pinned timings.

Stage	Default (s)	Pinned (s)	Default ÷ pinned
OAT's SIMP code, CG + OC, 256^2	47.145	37.043	1.27
OAT's SIMP code, CG + OC, 64^2	18.117	14.637	1.24
OAT's SIMP code, SPSOLVE + OC, 64^2	47.965	44.366	1.08
Our baseline, GPU, 256^2 (OAT problems)	9.698	8.450	1.15
Our baseline, GPU, 64^2 (OAT problems)	1.775	1.523	1.17
Our baseline, PARDISO, 64^2 (OAT problems)	0.777	0.607	1.28
Our baseline, PARDISO, 256^2 (OAT problems)	19.614	13.047	1.50
OAT inference, 256^2	0.533	0.471	1.13
OAT inference, 64^2	0.521	0.467	1.11
Our baseline, GPU, cantilever, 64^2 (benchmark)	2.378	2.015	1.18
Our baseline, PARDISO, cantilever, 64^2 (benchmark)	0.733	0.577	1.27
Our baseline, GPU, cantilever, 256^2 (benchmark)	4.615	4.343	1.06
Our baseline, PARDISO, cantilever, 256^2 (benchmark)	50.329	13.600	3.70
Our baseline, GPU, MBB beam, 64^2 (benchmark)	2.293	1.983	1.16
Our baseline, PARDISO, MBB beam, 64^2 (benchmark)	1.841	0.593	3.10
Our baseline, GPU, MBB beam, 256^2 (benchmark)	4.749	4.509	1.05
Our baseline, PARDISO, MBB beam, 256^2 (benchmark)	38.090	14.111	2.70

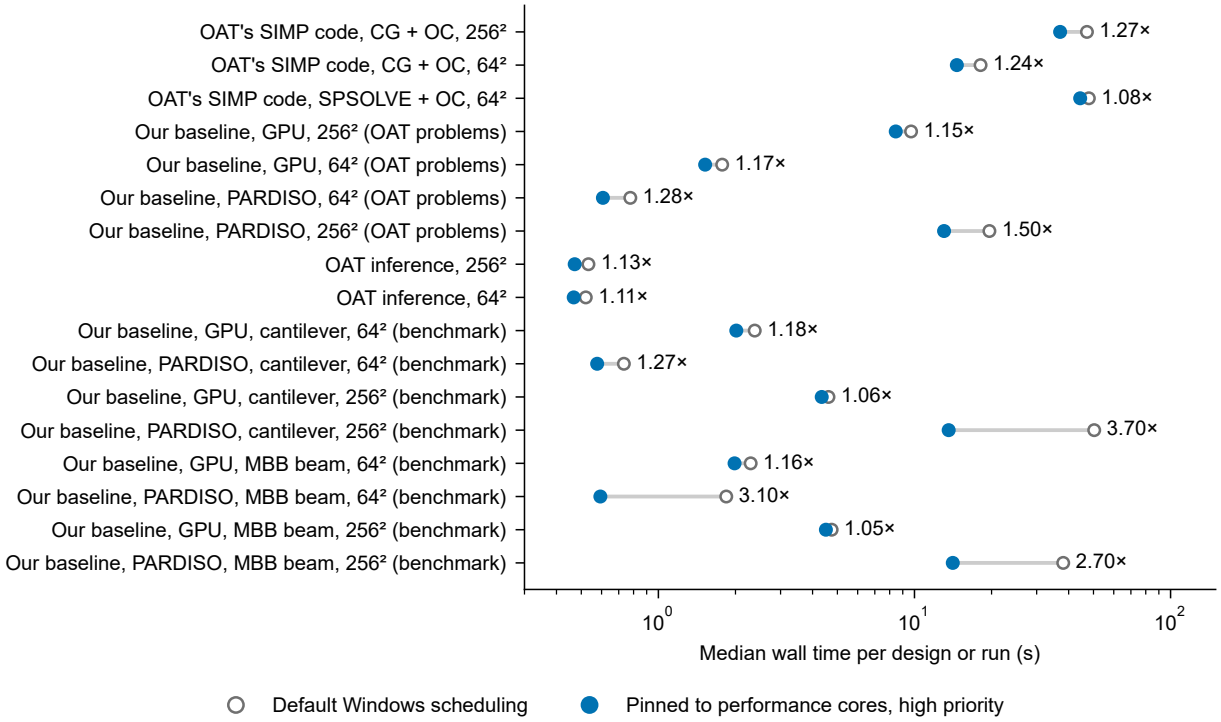


Fig. OR2.3 Median wall time under default scheduling (open) and pinned (filled), with the ratio. OAT, Optimize Any Topology, a diffusion model; GPU/CPU, graphics/central processing unit; CG+OC, conjugate gradient with an optimality-criteria update; SPSOLVE, the sparse direct-solver path. Source: `figures/data/ed9_scheduling.csv`.

5. Energy

Method: nvidia-smi board power, 100 ms, trapezoid; idle from a 5 s trace. Board energy integrates the sampled board power over the timed window; the above-idle value subtracts the idle power measured before each run.

Stage	Problems	Board energy per design (J, median)	Above idle (J, median)
OAT inference 256	5	102.4	67.2
OAT inference 64	5	100.9	65.1
ours GPU 256	5	1560.3	991.0
ours GPU 64	5	199.7	155.5

Stage	Problem	Board J	Above idle J	Idle W	Window (s)	Repeats
ours GPU 256	1015	630.5	379.9	65.5	11.47	3
ours GPU 256	1676	1949.3	1203.1	65.5	34.16	3
ours GPU 256	1839	2387.9	1522.2	65.5	39.63	3
ours GPU 256	1935	555.0	337.4	65.5	9.96	3
ours GPU 256	1961	1560.3	991.0	65.5	26.07	3
ours GPU 64	15	221.2	172.0	27.2	5.43	3
ours GPU 64	676	163.7	127.2	27.2	4.03	3
ours GPU 64	839	199.7	155.5	27.2	4.88	3
ours GPU 64	935	167.3	130.2	27.2	4.10	3
ours GPU 64	961	354.7	276.8	27.2	8.59	3
OAT inference 256	1015	102.4	67.5	76.2	4.59	10
OAT inference 256	1676	101.6	67.2	76.2	4.50	10
OAT inference 256	1839	100.8	66.5	76.2	4.51	10
OAT inference 256	1935	102.6	66.9	76.2	4.68	10
OAT inference 256	1961	104.0	69.9	76.2	4.48	10
OAT inference 64	15	98.0	62.4	77.2	4.61	10
OAT inference 64	676	105.3	69.2	77.2	4.67	10
OAT inference 64	839	97.2	62.8	77.2	4.45	10
OAT inference 64	935	104.1	68.7	77.2	4.59	10

Stage	Problem	Board J	Above idle J	Idle W	Window (s)	Repeats
OAT inference 64	961	100.9	65.1	77.2	4.64	10

6. OAT re-test: per-problem results

64^2 — compliance error (CE, % above the dataset reference) and volume (`oat_eval_n64.json`; designs after OAT inference, OAT's SIMP code and 150 iterations of our baseline).

Problem	Target volume	CE OAT	CE OAT-SIMP[CG+OC]	CE OAT-SIMP[direct solve+OC]	CE ours[gpu]	CE ours[pardiso]
15	0.300	20.22 %	3.22 %	3.18 %	3.05 %	3.05 %
676	0.460	-0.02 %	0.25 %	0.26 %	-0.00 %	-0.00 %
839	0.460	0.16 %	0.02 %	0.01 %	-0.01 %	-0.01 %
935	0.440	0.57 %	0.03 %	0.03 %	0.04 %	0.04 %
961	0.320	2.16 %	2.21 %	2.25 %	2.70 %	2.70 %

64^2 — wall time per design (s) (`oat_retest_summary.json`, pinned; median of the timed repeats). OAT's SIMP times and our loop times exclude setup; the last column(s) add our per-design setup (`ttq_summary.json`).

Problem	OAT SIMP 64 CG+OC	OAT SIMP 64 SP-SOLVE+OC	ours GPU 64	ours PARDISO 64	OAT inference 64	ours gpu, 150 its incl. setup	ours pardiso, 150 its incl. setup
15	27.1	46.4	1.74	0.591	0.464	2.11	1.49
676	9.78	43.5	1.23	0.607	0.468	1.5	1.51
839	14.3	44.4	1.52	0.638	0.486	1.89	1.55
935	14.6	46.2	1.27	0.631	0.459	1.59	1.54
961	38.5	44.3	2.79	0.607	0.467	3.37	1.52

256^2 — compliance error (CE, % above the dataset reference) and volume (`oat_eval_n256.json`; designs after OAT inference, OAT's SIMP code and 150 iterations of our baseline).

Problem	Target volume	CE OAT	CE OAT-SIMP[CG+OC]	CE ours[gpu]	CE ours[pardiso]
1015	0.380	-7.42 %	10.83 %	-8.19 %	-8.19 %
1676	0.300	0.28 %	2.32 %	0.47 %	0.47 %
1839	0.320	4.50 %	4.11 %	2.01 %	2.01 %
1935	0.300	-0.31 %	-0.20 %	-0.36 %	-0.36 %
1961	0.440	3.76 %	1.42 %	0.67 %	0.67 %

256^2 — wall time per design (s) (`oat_retest_summary.json`, pinned; median of the timed repeats). OAT's SIMP times and our loop times exclude setup; the last column(s) add our per-design setup (`ttq_summary.json`).

Problem	OAT SIMP 256 CG+OC	ours GPU 256	ours PARDISO 256	OAT inference 256	ours gpu, 150 its incl. setup	ours pardiso, 150 its incl. setup
1015	37.3	3.22	13	0.477	3.91	16.4
1676	37.2	9.8	13.2	0.502	10.1	17.9
1839	36.8	13.6	14.1	0.465	12.9	16.6
1935	20.7	2.74	13	0.471	2.95	17.1
1961	37	8.45	12.9	0.471	8.76	16.9

7. OAT re-test: matched quality under both rules

For each problem, our baseline's tested budgets (5, 10, 15, 20, 30, 40, 50, 75, 100 and 150 iterations) are scanned for the first budget whose CE is at or below OAT's. The pre-registered rule compares CE only.

The post hoc primary rule also requires that budget's volume not to exceed OAT's volume or the target, whichever is larger (Sect. 3.3.1 of the paper). Unmatched problems are censored at 150 iterations (`ttq_summary.json`).

64^2, gpu (OAT inference median 0.467 s)

Problem	CE OAT	Volume OAT	Target volume	Primary: matched	Iterations	Wall incl. setup (s)	Pre-reg.: matched	Iterations	Wall incl. setup (s)
15	20.22 %	0.301	0.300	yes	15	0.286	yes	15	0.286
676	-0.02 %	0.461	0.460	no	–	–	no	–	–
839	0.16 %	0.455	0.460	yes	5	0.171	yes	5	0.171
935	0.57 %	0.434	0.440	yes	10	0.229	yes	10	0.229
961	2.16 %	0.320	0.320	no	–	–	no	–	–

64^2, pardiso (OAT inference median 0.467 s)

Problem	CE OAT	Volume OAT	Target volume	Primary: matched	Iterations	Wall incl. setup (s)	Pre-reg.: matched	Iterations	Wall incl. setup (s)
15	20.22 %	0.301	0.300	yes	15	0.887	yes	15	0.887
676	-0.02 %	0.461	0.460	no	–	–	no	–	–
839	0.16 %	0.455	0.460	yes	5	0.824	yes	5	0.824
935	0.57 %	0.434	0.440	yes	10	0.86	yes	10	0.86
961	2.16 %	0.320	0.320	no	–	–	no	–	–

256^2, gpu (OAT inference median 0.471 s)

Problem	CE OAT	Volume OAT	Target volume	Primary: matched	Iterations	Wall incl. setup (s)	Pre-reg.: matched	Iterations	Wall incl. setup (s)
1015	-7.42 %	0.377	0.380	no	–	–	yes	5	0.232
1676	0.28 %	0.305	0.300	no	–	–	no	–	–
1839	4.50 %	0.318	0.320	yes	20	1.61	yes	20	1.61
1935	-0.31 %	0.300	0.300	yes	40	0.94	yes	5	0.276
1961	3.76 %	0.436	0.440	yes	15	0.812	yes	15	0.812

256^2, pardiso (OAT inference median 0.471 s)

Problem	CE OAT	Volume OAT	Target volume	Primary: matched	Iterations	Wall incl. setup (s)	Pre-reg.: matched	Iterations	Wall incl. setup (s)
1015	-7.42 %	0.377	0.380	no	–	–	yes	5	1.72
1676	0.28 %	0.305	0.300	no	–	–	no	–	–
1839	4.50 %	0.318	0.320	yes	20	3.35	yes	20	3.35
1935	-0.31 %	0.300	0.300	yes	40	5.11	yes	5	1.76
1961	3.76 %	0.436	0.440	yes	15	2.92	yes	15	2.92

Summary at 256^2 (GPU): under the pre-registered rule 4 of 5 problems matched, and the median time ratio (ours ÷ OAT) was 1.72; the corresponding median-input break-even number of uses is 46.6 million. At 64^2 with PARDISO the primary-rule ratio was 1.91.

8. OAT break-even note

Formula: $\tau = C_{\text{train}}/(C_{\text{classical}} - C_{\text{inference}})$, where τ is the repayment count and the C terms are offline cost and per-design times (OAT App. G). Training cost: 168 + 16 H100-days, as stated in OAT App. G (15,897,600 H100-seconds). OAT states a break-even of 2,320,000 uses, which implies a per-design saving of 6.85 s. Except the two rows labelled mixed hardware, each row below uses times from a single machine.

Scenario	Mesh	Classical time	s	Learned time	s	Saving (s)	Break-even uses
OAT Table 6 as reported	64^2	SIMP GPU	9.13	OAT	0.508	8.62	1.84 million
OAT Table 6 as reported	64^2	SIMP GPU	9.13	OAT w/ Ref	0.637	8.49	1.87 million
OAT Table 6 as reported	64^2	SIMP (CPU)	3.45	OAT	0.508	2.94	5.40 million
OAT Table 6 as reported	64^2	SIMP (CPU)	3.45	OAT w/ Ref	0.637	2.81	5.65 million
OAT Table 6 as reported	256^2	SIMP GPU	68.3	OAT	0.51	67.8	0.23 million
OAT Table 6 as reported	256^2	SIMP GPU	68.3	OAT w/ Ref	3.28	65	0.24 million
OAT Table 6 as reported	256^2	SIMP (CPU)	69.5	OAT	0.51	68.9	0.23 million
OAT Table 6 as reported	256^2	SIMP (CPU)	69.5	OAT w/ Ref	3.28	66.2	0.24 million
same machine, 150 iterations, loop only (pre-registered timing)	64^2	ours GPU (median of 5)	1.52	OAT inference, no Ref (median of 5)	0.467	1.06	15.06 million
same machine, 150 iterations, loop only (pre-registered timing)	64^2	ours PARDISO CPU (median of 5)	0.607	OAT inference, no Ref (median of 5)	0.467	0.14	113.39 million
ours vs OAT w/ Ref as reported (mixed hardware)	64^2	ours GPU (median of 5)	1.52	OAT w/ Ref (Table 6)	0.637	0.886	17.95 million
same machine, 150 iterations, loop only (pre-registered timing)	256^2	ours GPU (median of 5)	8.45	OAT inference, no Ref (median of 5)	0.471	7.98	1.99 million
same machine, 150 iterations, loop only (pre-registered timing)	256^2	ours PARDISO CPU (median of 5)	13	OAT inference, no Ref (median of 5)	0.471	12.6	1.26 million
ours vs OAT w/ Ref as reported (mixed hardware)	256^2	ours GPU (median of 5)	8.45	OAT w/ Ref (Table 6)	3.28	5.17	3.07 million
same machine, 150 iterations incl. setup	256^2	ours gpu, 150 its incl. setup (median of 5)	8.76	OAT inference, no Ref (median of 5)	0.471	8.29	1.92 million
same machine, matched quality (primary, censored median ratio)	256^2	ours gpu: median ratio 3.466 x OAT time (3/5 matched)	1.63	OAT inference, no Ref (median of 5)	0.471	1.16	13.68 million
same machine, matched quality (primary, matched problems only)	256^2	ours gpu, median over 3/5 matched problems	0.94	OAT inference, no Ref (median of 5)	0.471	0.469	33.89 million
same machine, matched quality (pre-registered rule, censored median ratio)	256^2	ours gpu: median ratio 1.724 x OAT time (4/5 matched)	0.812	OAT inference, no Ref (median of 5)	0.471	0.341	46.60 million
same machine, matched quality (pre-registered rule, matched problems only)	256^2	ours gpu, median over 4/5 matched problems	0.544	OAT inference, no Ref (median of 5)	0.471	0.073	217.81 million
same machine, 150 iterations incl. setup	64^2	ours gpu, 150 its incl. setup (median of 5)	1.89	OAT inference, no Ref (median of 5)	0.467	1.43	11.14 million
same machine, matched quality (primary, censored median ratio)	64^2	ours gpu: median ratio 0.617 x OAT time (3/5 matched)	0.288	OAT inference, no Ref (median of 5)	0.467	-0.179	never (classical faster)
same machine, matched quality (primary, matched problems only)	64^2	ours gpu, median over 3/5 matched problems	0.229	OAT inference, no Ref (median of 5)	0.467	-0.238	never (classical faster)
same machine, matched quality (pre-registered rule, censored median ratio)	64^2	ours gpu: median ratio 0.617 x OAT time (3/5 matched)	0.288	OAT inference, no Ref (median of 5)	0.467	-0.179	never (classical faster)
same machine, matched quality (pre-registered rule, matched problems only)	64^2	ours gpu, median over 3/5 matched problems	0.229	OAT inference, no Ref (median of 5)	0.467	-0.238	never (classical faster)
same machine, 150 iterations incl. setup	64^2	ours pardiso, 150 its incl. setup (median of 5)	1.52	OAT inference, no Ref (median of 5)	0.467	1.05	15.08 million
same machine, matched quality (primary, censored median ratio)	64^2	ours pardiso: median ratio 1.912 x OAT time (3/5 matched)	0.893	OAT inference, no Ref (median of 5)	0.467	0.426	37.33 million
same machine, matched quality (primary, matched problems only)	64^2	ours pardiso, median over 3/5 matched problems	0.86	OAT inference, no Ref (median of 5)	0.467	0.393	40.48 million
same machine, matched quality (pre-registered rule, censored median ratio)	64^2	ours pardiso: median ratio 1.912 x OAT time (3/5 matched)	0.893	OAT inference, no Ref (median of 5)	0.467	0.426	37.33 million

Scenario	Mesh	Classical time	s	Learned time	s	Saving (s)	Break-even uses
same machine, matched quality (pre-registered rule, matched problems only)	64^2	ours pardiso, median over 3/5 matched problems	0.86	OAT inference, no Ref (median of 5)	0.467	0.393	40.48 million
same machine, 150 iterations incl. setup	256^2	ours pardiso, 150 its incl. setup (median of 5)	16.9	OAT inference, no Ref (median of 5)	0.471	16.4	0.97 million
same machine, matched quality (primary, censored median ratio)	256^2	ours pardiso: median ratio 10.845 x OAT time (3/5 matched)	5.11	OAT inference, no Ref (median of 5)	0.471	4.64	3.43 million
same machine, matched quality (primary, matched problems only)	256^2	ours pardiso, median over 3/5 matched problems	3.35	OAT inference, no Ref (median of 5)	0.471	2.87	5.53 million
same machine, matched quality (pre-registered rule, censored median ratio)	256^2	ours pardiso: median ratio 6.208 x OAT time (4/5 matched)	2.93	OAT inference, no Ref (median of 5)	0.471	2.45	6.48 million
same machine, matched quality (pre-registered rule, matched problems only)	256^2	ours pardiso, median over 4/5 matched problems	2.34	OAT inference, no Ref (median of 5)	0.471	1.87	8.50 million

Every row above uses a median. A workload repays the offline cost through its mean per-design saving. Counting each unmatched problem at its 150-iteration time, which is a lower bound on its classical time, the five-problem mean saving with our GPU baseline gives a break-even of at most 5.3 million uses at 256^2 (7.5 million under the pre-registered rule) and at most 24.7 million at 64^2 under either rule (Sect. 4.5 of the paper).

9. HGTO re-test: all six configurations

Hardware note: same machine: RTX 5090, i5-14600KF (6 P-cores), pinned; article: RTX 6000 Ada, Threadripper PRO 5995WX with two threads; published and measured values are never mixed in one ratio.

Equivalence gate (`hgto_gate.json`): our GPU port of the authors' SIMP–OC against their two-thread code.

Problem	Updates compared	Max. relative difference	Pass
cantilever 120×40	50	3.92×10^{-11}	yes
MBB half-beam	50	8.03×10^{-11}	yes
inclined-load cantilever	50	3.55×10^{-11}	yes

Problem	Compliance (authors' code)	Compliance (port)	Relative difference	Updates (authors' / port)	Pass
cantilever 120×40	175.082	175.082	4.79×10^{-11}	298 / 298	yes
MBB half-beam	191.493	191.493	7.29×10^{-11}	292 / 292	yes
inclined-load cantilever	132.722	132.722	4.95×10^{-11}	1000 / 1000	yes
cantilever 240×80	175.765	175.765	1.14×10^{-11}	208 / 208	yes
cantilever 480×160	171.584	171.584	2.52×10^{-11}	291 / 291	yes
cantilever 960×320	168.555	168.555	4.60×10^{-11}	571 / 571	yes

Wall times (s) and final compliance (`hgto_retest.json`). Published times are the article's own, on its hardware; measured times are medians on our machine, with the number of repeats in brackets. OC-2 and OC-6: the authors' SIMP–OC with two and six CPU threads; port: our GPU port.

Configuration	Published HGTO	Published OC	HGTO	OC-2	OC-6	Port	Compliance HGTO	Compliance port
cantilever 120×40	8.27	9.02	10 (3)	9.95 (3)	9.28 (3)	7.22 (3)	176.49	175.08
MBB half-beam	7.43	8.99	9.14 (3)	9.95 (3)	9.12 (3)	8.34 (3)	195.47	191.49

Configuration	Published HGTO	Published OC	HGTO	OC-2	OC-6	Port	Compliance HGTO	Compliance port
inclined-load cantilever	8.49	26.9	10.5 (3)	29.4 (3)	26.9 (3)	29.1 (3)	133.48	132.72
cantilever 240×80	11.7	30.9	15.3 (3)	39.3 (3)	37.1 (3)	7.15 (3)	176.85	175.77
cantilever 480×160	47.4	198	49.2 (3)	243 (3)	219 (3)	19.2 (3)	172.44	171.58
cantilever 960×320	196	1,880	197 (3)	2,141 (1)	1,876 (1)	106 (3)	171	168.56

Configuration	HGTO ÷ port (to convergence)	HGTO ÷ port (matched quality)	Port termination (updates)	HGTO termination (updates)
cantilever 120×40	1.39	2.39	converged (298)	converged (283)
MBB half-beam	1.10	1.98	converged (292)	converged (281)
inclined-load cantilever	0.36	2.27	iteration cap (1000)	converged (294)
cantilever 240×80	2.14	2.77	converged (208)	converged (258)
cantilever 480×160	2.57	4.72	converged (291)	converged (312)
cantilever 960×320	1.85	6.51	converged (571)	converged (345)

10. OAT re-test: post hoc extension to 25 problems and five OAT samples

Post hoc (baseline pre-registration §9, row of 2026-09-25). Per size, the five primary problems plus 20 problems drawn from the same test split with seed 20260925, disjoint from the five; five OAT sampling seeds per problem. Each OAT sample is scored with OAT's evaluator and matched by the first budget of our GPU baseline under both rules; the ratio is our setup-inclusive time at that budget over that sample's inference time. Per problem, the median over the five samples is given; 'unmatched' means that at least three of five samples were not matched within 150 iterations. Sign test: exact, two-sided, over problems; it counts which method is faster in each paired problem (Dixon and Mood, 1946) (`oat_extended.json`).

64^2 (run `retest/runs/20260925T070716Z_oat_extended_n64`): OAT inference 0.450 s median (0.443–0.714 s over all samples); median OAT CE 0.40 %; median range of OAT CE across seeds 1.34 percentage points.

Problem	Primary five	Median ratio, post hoc rule	Median ratio, pre-registered rule
15	yes	0.62	0.62
676	yes	0.42	0.42
839	yes	0.95	0.95
935	yes	0.48	0.48
961	yes	1.53	1.53
86		0.31	0.31
121		1.42	1.42
134		2.02	2.02
241		unmatched	unmatched
338		0.43	0.43
403		unmatched	unmatched
443		0.33	0.33
499		0.31	0.31
523		0.40	0.40
530		1.51	1.51
580		0.76	0.75
581		0.46	0.46
629		0.89	0.67
681		unmatched	unmatched
760		0.69	0.69
788		0.63	0.63
817		unmatched	unmatched
972		unmatched	unmatched
974		0.40	0.40
985		1.81	1.81

Rule	Median ratio (25 problems)	Primary five	New twenty	Baseline faster	Sign test p	Samples matched
post hoc (CE and volume)	0.76	0.62	0.83	15 of 25	0.424	90 of 125
pre-registered (CE only)	0.69	0.62	0.72	15 of 25	0.424	92 of 125

256^2 (run `retest/runs/20260925T071400Z_oat_extended_n256`): OAT inference 0.478 s median (0.448–0.607 s over all samples); median OAT CE 0.58 %; median range of OAT CE across seeds 1.41 percentage points.

Problem	Primary five	Median ratio, post hoc rule	Median ratio, pre-registered rule
1015	yes	unmatched	0.45
1676	yes	unmatched	unmatched
1839	yes	2.93	2.93
1935	yes	0.81	0.54
1961	yes	1.59	1.59
1086		1.00	1.00
1121		2.63	2.63
1134		0.50	0.50
1241		8.34	8.34
1338		unmatched	unmatched
1403		1.06	1.06
1443		0.64	0.64
1499		1.36	1.36
1523		4.68	4.68
1530		unmatched	unmatched
1580		unmatched	unmatched
1581		0.48	0.48
1629		0.85	0.85
1681		1.35	1.33
1760		unmatched	unmatched
1788		1.77	1.77
1817		1.47	1.47
1972		1.43	1.43
1974		2.18	1.67
1985		0.87	0.87

Rule	Median ratio (25 problems)	Primary five	New twenty	Baseline faster	Sign test p	Samples matched
post hoc (CE and volume)	1.59	2.93	1.45	7 of 25	0.0433	90 of 125
pre-registered (CE only)	1.47	1.59	1.45	8 of 25	0.108	96 of 125